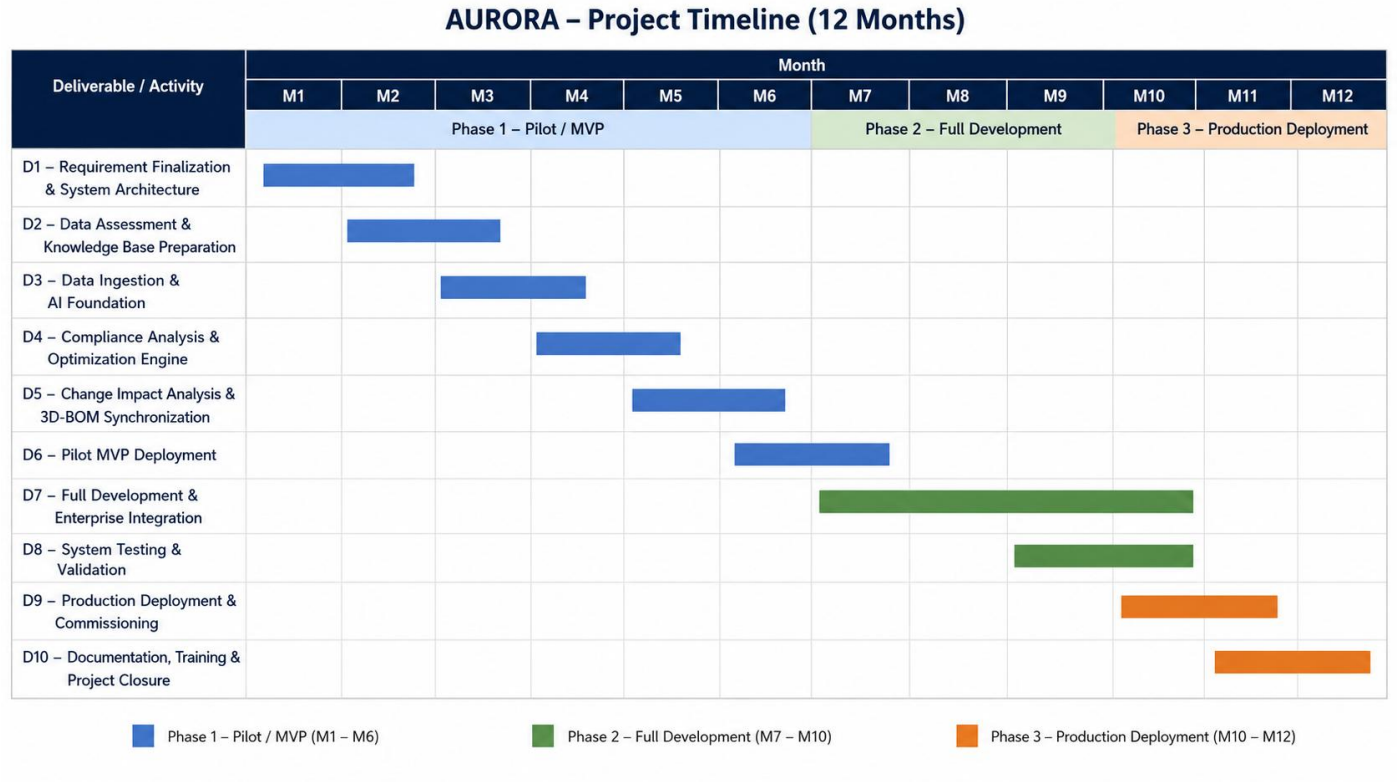
DETAILED PROJECT TIMELINE – AURORA

AI Tool for Design Optimization with Class Rule Compliance & 3D Model–Bill of Material (BOM) Synchronization

Total Duration: 12 Months

Project Reference: MDL DRISHTI Innovation Challenge – Problem Statement 71

The project shall be executed in accordance with the Delivery Phases defined in the Scope of Work. The implementation roadmap covers capability demonstration, pilot deployment, full system development and secure production deployment while ensuring compliance with all functional and non-functional requirements specified by MDL.



PHASE 0 – CAPABILITY DEMONSTRATION (Before Project Start)

Objectives

Demonstrate the feasibility of the proposed AI platform by showcasing automated compliance analysis and/or 3D–BOM synchronization using representative engineering datasets.

Activities

- Demonstration of CAD model ingestion
- Demonstration of engineering document ingestion
- AI-assisted class rule compliance checking
- Automated clause citation
- 3D–BOM synchronization demonstration
- Sample compliance report generation
- Explainability demonstration
- Offline deployment demonstration

Outputs

- Capability Demonstration Report
- Demonstration Video

• Technical Feasibility Assessment
• Demonstration Feedback Report

PHASE 1 – PILOT / MVP

(Month 0 – Month 6)

MONTH 0–2

MILESTONE M1

DELIVERABLE D1 – REQUIREMENT FINALIZATION & SOLUTION ARCHITECTURE

Objectives

Establish the project scope, identify stakeholders, define the system architecture, assess available engineering data and finalize the implementation roadmap.

Activities

• Project Kick-off Meeting with MDL
• Requirement Elicitation Workshops
• Review of Existing Design Workflows
• Review of CAD and BOM Management Practices
• Identification of Applicable Classification Rules
• Identification of Standards and Regulations
• Definition of Pilot Scope
• Security Requirement Analysis
• Air-Gapped Deployment Planning
• Preparation of Software Requirement Specification (SRS)
• High-Level Architecture Design
• Initial Risk Assessment

Outputs
• Software Requirement Specification
• Solution Architecture Document
• Project Management Plan
• Stakeholder Matrix
• Data Requirement Matrix
• Security Requirement Specification

Review Gate RG-1

Approval of Project Scope, Architecture and Pilot Success Criteria.

MONTH 2–4

MILESTONE M2

DELIVERABLE D2 – DATA INGESTION & KNOWLEDGE BASE

Activities
• Collection of CAD Models
• Collection of BOM Data

• Collection of Engineering Drawings
• Collection of Class Rules
• Collection of Standards
• OCR Processing Pipeline
• Metadata Extraction
• Semantic Indexing
• Knowledge Base Construction
• Version Control Framework

Outputs
• Engineering Knowledge Base
• Data Ingestion Framework
• Rule Repository
• Document Processing Pipeline

MONTH 4–6

MILESTONE M3

DELIVERABLE D3 – PILOT AI PLATFORM

Activities
• Compliance Analysis Engine
• Rule Citation Engine
• Engineering Optimization Suggestions
• 3D–BOM Matching
• Mismatch Detection
• Traceability Framework
• Pilot Dashboard
• Pilot Reporting
• Internal Validation

Outputs
• Pilot AI Platform
• Compliance Analysis Module
• 3D–BOM Synchronization Engine
• Pilot Demonstration

Review Gate RG-2

Pilot Acceptance Review.

PHASE 2 – FULL DEVELOPMENT

(Month 6 – Month 10)

MONTH 6–8

MILESTONE M4

DELIVERABLE D4 – ADVANCED AI ENGINE

Activities

• Multi-disciplinary Rule Engine
• Cross-discipline Impact Analysis
• Structural Impact Prediction
• Weight Redistribution Analysis
• Revision Management
• Workflow Engine
• Review & Approval Workflow
• Explainability Framework

Outputs
• Advanced AI Engine
• Change Impact Module
• Revision Workflow
• Review Dashboard

MONTH 8–10

MILESTONE M5

DELIVERABLE D5 – ENTERPRISE PLATFORM

Activities
• PLM Integration
• CAD Integration
• ERP Integration
• REST APIs
• RBAC
• Audit Logging
• Analytics Dashboard
• PDF Report Generation
• Performance Optimization

Outputs
• Enterprise Platform
• Integration Framework
• Reporting Engine
• Analytics Dashboard

Review Gate RG-3

Full Functional Review and Enterprise Acceptance.

PHASE 3 – AIR-GAPPED PRODUCTION DEPLOYMENT

(Month 10 – Month 12)

MONTH 10–11

MILESTONE M6

DELIVERABLE D6 – TESTING & VALIDATION

Activities
• Functional Testing

• Integration Testing
• Security Testing
• Air-Gapped Validation
• User Acceptance Testing
• Performance Benchmarking
• Cyber Security Validation
• Defect Resolution

Outputs
• Testing Report
• Validation Report
• UAT Sign-Off
• Security Compliance Report

MONTH 11–12

MILESTONE M7

DELIVERABLE D7 – PRODUCTION DEPLOYMENT & PROJECT CLOSURE

Activities

• Production Deployment
• Air-Gapped Installation
• Administrator Training
• User Training
• Documentation Handover
• Operational Readiness Assessment
• Final Demonstration
• Knowledge Transfer
• Project Closure

Outputs
• Production Deployment
• Administrator Manual
• User Manual
• Training Report
• Operational Readiness Report
• Project Closure Report

Review Gate RG-4

Final Acceptance and Commissioning Approval.

EXECUTIVE TIMELINE SUMMARY

Month	Milestone	Deliverable
M0–M2	M1	D1 – Requirement Finalization & Solution Architecture
M2–M4	M2	D2 – Data Ingestion & Knowledge Base
M4–M6	M3	D3 – Pilot AI Platform
M6–M8	M4	D4 – Advanced AI Engine
M8–M10	M5	D5 – Enterprise Platform
M10–M11	M6	D6 – Testing & Validation
M11–M12	M7	D7 – Production Deployment & Project Closure

MILESTONE SUMMARY

Milestone	Key Outcome
M1	Project Initiated, Requirements Finalized and Architecture Approved
M2	Engineering Data Ingestion Framework and Knowledge Base Completed
M3	Pilot AI Platform Successfully Demonstrated
M4	Advanced Compliance and Impact Analysis Engine Completed
M5	Enterprise Platform and Integrations Completed
M6	Testing, Validation and User Acceptance Completed
M7	Air-Gapped Production Deployment, Training and Project Closure Completed